

Henri Stewart Wilks-Stebbing

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Doctoral researcher in computational geomorphology, quantifying how climate change is reorganising the thermal, geomorphological and biological systems of the Eastern Alps. Building automated Python workflows on Linux HPC and Google Earth Engine to process multi-decadal climate reanalysis, DEM and satellite time series at regional scale, and apply non-parametric trend analysis and process-based frost weathering models to identify where landscapes are most sensitive to warming. Working closely with collaborators in statistics and machine learning to connect these signals to slope stability and ecosystem change.

Education

Ph.D. (Dr. rer. nat.) in Geology

University of Salzburg, Austria

expected Mar. 2029

M.Sc. in Earth Sciences

Vrije Universiteit Amsterdam, Netherlands

Jun. 2025

B.Sc. in Physical Geography

Liverpool Hope University, England

Jun. 2023

Experience

Doctoral Researcher

University of Salzburg, Division of Geology and Physical Geography

Salzburg, Austria

Oct. 2025 – Present

- Investigating the influence of climate change on landscape evolution in the Eastern Alps using Big Data workflows to facilitate large-scale space-time analyses

Research Assistant, Geomorphology and Quaternary Sedimentology Group

Vrije Universiteit Amsterdam, Department of Earth Science

Amsterdam, Netherlands

Aug. 2024 – Apr. 2025

- Led remote-sensing data acquisition for post-glacial landslide research in New Zealand
- Collaborated within an international, multidisciplinary team of scientists (New Zealand, UK, Netherlands)

Skills

Computation: Linux HPC and cluster computing (SLURM), cloud geoprocessing (Google Earth Engine), large-raster and big data processing

Programming: Python (xarray, GDAL), Bash, GMT, MATLAB, JavaScript (Earth Engine), $\text{\LaTeX}$

Analytical & Field: DEM and terrain analysis, non-parametric trend analysis (Mann-Kendall), frost weathering modelling, frequency-magnitude scaling, landslide inventory mapping, geomorphological mapping, remote sensing, XRF

Communication: Doctoral representative, scientific writing, oral presentation, workshop organisation, science communication

Publications

Manuscripts in preparation

Towards a Stress-Based Stability Criterion for Rock Slopes From 3D Stress Modeling of Entire Mountain Massifs (Haunsperger, V., Robl, J., Hergarten, S., Argentin, A.-L., **Wilks-Stebbing, H.**, Schröder, A., in prep)

Hidden Erosion: Quantifying Carbonate Mass Loss in the Taugl Catchment, Northern Calcareous Alps (Pleyer, B., Robl, J., Dremel, F., **Wilks-Stebbing, H.**, Tenczer, V., Otto, J.C., Hilberg, S., Rinder, T., Hauzenberger, C., in prep)

Tectonic and Climatic Controls on Valley-Floor Width Along the Alpine Fault, New Zealand (Clubb, F.J., Roda-Boluda, D.C., **Wilks-Stebbing, H.**, de Vries, L., in prep)

Conference contributions

Wilks-Stebbing, H., Robl, J., Meier, W., Tribsch, A., Trutschnig, W., Hergarten, S. **Spatiotemporal Gradients of Environmental Forcing across the Eastern Alps: Quantifying Drivers of Landscape and Ecosystem Dynamics.** *EGU General Assembly 2026, Vienna, EGU26-5449*, doi:10.5194/egusphere-egu26-5449.

Haunsperger, V., Robl, J., Hergarten, S., Argentin, A.-L., **Wilks-Stebbing, H.**, Schröder, A. **Towards a Stress-Based Stability Criterion for Rock Slopes From 3D Stress Modeling of Entire Mountain Massifs.** *EGU General Assembly 2026, Vienna, EGU26-5585*, doi:10.5194/egusphere-egu26-5585.

Clubb, F.J., Roda-Boluda, D.C., **Wilks-Stebbing, H.**, de Vries, L. **Tectonic and Climatic Controls on Valley-Floor Width Along the Alpine Fault, New Zealand.** *AGU Fall Meeting 2024, Washington, D.C., oral presentation by F. Clubb*, abstract U31B-05.

Peer Review

Nature Communications, early-career co-reviewer (1 manuscript)

2026

Past and Ongoing Research Projects

Landscape Response to a Changing Climate in the Eastern Alps: Analysing Big Data for Past, Present, and Future Conditions

(PhD Dissertation)

Oct. 2025 – Present

- Quantifying multi-decadal trends in frost cracking, freeze-thaw cycling and growing-season length across the Eastern Alps from daily gridded climate data (CHELSA-W5E5) and DEMs
- Testing coupling between thermal and vegetation change using MODIS NDVI time series in Google Earth Engine
- Determining changes in the erosional potential of Alpine torrents
- Contributing process-based inputs to a supervised deep learning approach for identifying hillslopes near critical failure

Constraining Landslide Rates, Controls, and Spatial Distributions in Calabria, Southern Italy

(MSc Thesis)

Jun. 2025

- Student collaborator through MSc Thesis on the RELICT project: examining how sediment dynamics and inherited relief impact landscape response to tectonic perturbations
- My role in this project involved creating multi-year landslide inventories which refined hazard assessment and landslide erosion rate estimates

Landslide Frequency and Magnitude in the Southern Alps, New Zealand

(MSc Research Project)

Dec. 2024

- Independent 7-month research project delivering a novel multi-year landslide inventory with >2,700 landslides
- Implemented advanced geomorphological and numerical modelling techniques to analyse landslide frequency–magnitude trends
- Co-author of a talk presented at the American Geophysical Union Fall Meeting 2024 and of the future publication in preparation, to which the landslide data I generated is incorporated

The Implications of Climate Change on High Geomorphology and Paraglacial Slope Failures

(BSc Thesis)

Jun. 2023

- Investigated the impact of climate on hillslope processes, analysing mass movement frequency–magnitude
- Assessed geological properties, slope mechanics, and climate forcing to identify key drivers of instability
- Conducted systematic literature review to synthesise current research on slope failure mechanisms

Professional Development

Member, Doctoral Mentoring Committee (AlpsChange)

Jan. 2026 – Present

- Co-managing a gender-balanced advocacy body to support 20+ PhD students in conflict resolution and career planning
- Serving as a primary point of contact for students to address institutional or supervisory challenges, ensuring impartial peer support and programme transparency

Landslide Mapping “Hackathon”

May 2025

- Organised and led a one-day landslide mapping event for undergraduate students

- Delivered hands-on instruction in landslide delineation techniques and volume-area scaling calculations
- Served as a primary point of contact for participants, providing technical guidance and support

Awards

- Awarded the **Liverpool Geological Society Prize** as Best Performing Liverpool Hope Student in Physical Geography, 2023

References

Dr. Jörg Robl

Associate Professor
Universität Salzburg
joerg.robl@plus.ac.at

Dr. Duna Roda-Boluda

Assistant Professor
Universitat Autònoma de Barcelona
duna.roda@uab.cat

Dr. Benjamin Campforts

Assistant Professor
Vrije Universiteit Amsterdam
b.campforts@vu.nl